



INTERNSHIP REPORT

ARTIFICIAL INTELLIGENCE

**Submitted in Partial Fulfilment of the Requirements for the Award of Degree of
Bachelor of Technology (Computer Science & Engineering)**

Submitted By -

Name: Dev Vishwakarma

Roll Number: 25SCS1003004916

Semester: III

Submitted To -

Department of Computer Science & Engineering

IILM University

Internship Organization

Codec Technologies Pvt. Ltd.

Duration: 08 August 2026 – 07 September 2026

Mode: Online

Designation: Artificial Intelligence Intern

CERTIFICATE

The attached certificate confirms the successful completion of a one-month Artificial Intelligence Internship at Codec Technologies Pvt. Ltd. from **08 August 2026 to 07 September 2026** as an **Artificial Intelligence Intern**.



OFFER LETTER

The offer letter issued by Codec Technologies Pvt. Ltd. appointed me as an **Artificial Intelligence Intern** for a duration of one month from **08 August 2026 to 07 September 2026**, with training focused on industry-oriented Artificial Intelligence concepts and projects.



INTERNSHIP OFFER LETTER

Dear Dev Vishwakarma,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 08/08/2026 to 07/09/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

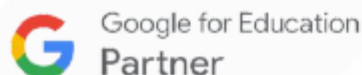
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 08/08/2026




Dr. Anurag Shrivastava
Talent Acquisition Manager



DECLARATION

I hereby declare that this internship report entitled "**Artificial Intelligence Internship Report**" submitted to the Department of Computer Science & Engineering, IILM University, is a genuine record of the internship work completed by me at **Codec Technologies Pvt. Ltd.**

I further declare that this report has been prepared based on my learning, practical activities, and projects completed during the internship period. The contents presented in this report are original and have not been submitted previously for the award of any degree, diploma, or certificate.

Any information obtained from books, journals, websites, or other sources has been properly acknowledged in the references section.

Submitted By -

Dev Vishwakarma

Roll No. 25SCS1003004916

B.Tech. (Computer Science & Engineering)

Semester III

Date: 10/09/2026

ACKNOWLEDGEMENT

The successful completion of this internship would not have been possible without the support and encouragement of many individuals and organizations.

I express my sincere gratitude to **Codec Technologies Pvt. Ltd.** for providing me with the opportunity to undertake a one-month internship in the field of Artificial Intelligence. This internship enabled me to strengthen my understanding of modern AI technologies, machine learning concepts, deep learning fundamentals, and practical implementation using Python.

I would also like to thank the mentors and training team of Codec Technologies for preparing well-structured learning modules and practical assignments throughout the internship period. Their guidance helped me understand various concepts including data preprocessing, supervised learning, neural networks, convolutional neural networks, natural language processing, and reinforcement learning.

I am equally grateful to the faculty members of **the Department of Computer Science & Engineering, IILM University**, for motivating students to participate in industrial training programs and gain practical exposure alongside academic studies.

Finally, I express my heartfelt gratitude to my parents and friends for their constant motivation, encouragement, and support throughout my internship.

Dev Vishwakarma

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CHAPTER 1

COMPANY PROFILE

1.1 Introduction

Codec Technologies Pvt. Ltd. is an educational technology and IT consultancy organization that provides internship programs, industrial training, technical workshops, and skill development courses for students and fresh graduates. The organization aims to bridge the gap between academic learning and industry requirements by offering practical exposure in various technical domains.

The company delivers training in several emerging technologies such as Artificial Intelligence, Machine Learning, Data Science, Web Development, Cloud Computing, Cyber Security, and Software Development. Through structured learning modules and project-based training, students gain practical knowledge that complements their academic curriculum.

The internship program is designed to provide industry-oriented learning through video lectures, assignments, quizzes, and project implementation. Students are encouraged to apply theoretical concepts to real-world problems, thereby improving both technical and analytical skills.

1.2 Vision

To empower students with practical technical knowledge and prepare them for successful careers in emerging technologies through industry-focused education and project-based learning.

1.3 Mission

- To provide high-quality technical education.
- To bridge the gap between academic knowledge and industrial requirements.
- To encourage innovation and practical problem-solving.
- To enhance students' employability through project-based internships.

1.4 Services Offered

Codec Technologies offers internships and training programs in multiple domains including:

- Artificial Intelligence
- Machine Learning

- Data Science
- Python Programming
- Web Development
- Java Development
- Cloud Computing
- Cyber Security
- Digital Marketing
- Software Development

Apart from internships, the organization also conducts workshops, certification programs, and industrial training sessions aimed at improving students' professional skills.

1.5 Internship Overview

During this one-month online internship, I completed structured learning modules covering Artificial Intelligence and Machine Learning fundamentals. The internship included theoretical concepts, practical implementation using Python, quizzes, and project-based learning.

The training modules focused on:

- Introduction to Artificial Intelligence
- Python Programming for AI
- Data Preprocessing
- Machine Learning Concepts
- Model Evaluation
- Deep Learning
- Convolutional Neural Networks (CNNs)
- Natural Language Processing (NLP)
- Reinforcement Learning
- AI Project Development

These modules helped me develop a better understanding of Artificial Intelligence and its real-world applications.

CHAPTER 2

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is one of the most significant technological advancements of the modern era. It refers to the ability of machines and computer systems to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, speech recognition, image analysis, and language understanding.

AI combines concepts from computer science, mathematics, statistics, and cognitive science to build intelligent systems capable of solving complex problems efficiently. Today, Artificial Intelligence is widely used in healthcare, finance, transportation, education, agriculture, cybersecurity, robotics, and many other industries.

The rapid growth of AI has transformed the way organizations analyze data, automate processes, and provide intelligent solutions to users. Machine learning algorithms enable systems to improve their performance through experience, while deep learning techniques allow computers to process large amounts of structured and unstructured data with remarkable accuracy.

During this internship, I explored the fundamental principles of Artificial Intelligence through structured learning modules provided by Codec Technologies. I gained practical exposure to Python programming, machine learning concepts, convolutional neural networks, natural language processing, and reinforcement learning. The internship also introduced me to real-world AI applications and project development methodologies.

The knowledge gained during this internship has strengthened my technical foundation and enhanced my understanding of modern Artificial Intelligence technologies, motivating me to continue learning advanced AI concepts in the future.

CHAPTER 3

OBJECTIVES OF THE INTERNSHIP

The primary objectives of this internship were:

- To understand the basic concepts of Artificial Intelligence.
- To learn Python programming for AI development.
- To understand the importance of data preprocessing and visualization.
- To study supervised machine learning algorithms.
- To learn techniques for evaluating machine learning models.
- To understand the working principles of deep learning and neural networks.
- To gain knowledge of Convolutional Neural Networks (CNNs) for image classification.
- To explore Natural Language Processing (NLP) techniques.
- To understand the basics of Reinforcement Learning.
- To apply theoretical knowledge by implementing practical AI projects.
- To improve problem-solving and analytical thinking skills.
- To gain exposure to industry-oriented AI tools and technologies.

CHAPTER 4

INTERNSHIP LEARNING MODULES

The Artificial Intelligence Internship conducted by **Codec Technologies** was divided into four structured learning weeks. Each week focused on a specific domain of Artificial Intelligence, gradually progressing from basic concepts to practical implementation of modern AI techniques. The learning approach combined theoretical understanding with practical demonstrations, enabling interns to understand how Artificial Intelligence is applied in real-world scenarios.

Throughout the internship, I studied concepts using video lectures, practical demonstrations, and quizzes. The structured curriculum helped me develop both conceptual knowledge and practical implementation skills in Artificial Intelligence.

4.1 Week 1 – Foundations of Artificial Intelligence and Python

The first week introduced the fundamentals of Artificial Intelligence and the programming language used extensively for AI development.

Initially, I learned the definition of Artificial Intelligence, its history, evolution, and importance in today's technology-driven world. Different branches of AI such as **Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Robotics** were introduced. Their real-life applications in healthcare, education, transportation, banking, and entertainment were also discussed.

Another major focus during this week was **Python programming**. Python is considered one of the most suitable programming languages for Artificial Intelligence because of its simple syntax and extensive collection of libraries. During this module, I became familiar with Python syntax, variables, loops, conditional statements, functions, lists, dictionaries, tuples, and object-oriented programming concepts.

The final module of Week 1 focused on **Data Preprocessing** and **Data Visualization**. Before any AI model can be trained, raw data needs to be cleaned and organized. I learned techniques for handling missing values, duplicate records, inconsistent data formats, and feature selection. Data visualization was performed using graphical techniques that help understand patterns and relationships within datasets.

This week established the basic foundation required for understanding advanced Artificial Intelligence concepts.

Learning Outcomes of Week 1

- Introduction to Artificial Intelligence
- Understanding AI applications
- Python programming fundamentals
- Data preprocessing techniques
- Data visualization concepts
- Importance of clean datasets

4.2 Week 2 – Machine Learning Basics

The second week focused on **Machine Learning**, one of the most important branches of Artificial Intelligence.

Machine Learning enables computers to learn from historical data and make predictions without being explicitly programmed. Instead of writing specific rules for every task, algorithms identify patterns in data and use them to make intelligent decisions.

The internship introduced Supervised Learning, where models are trained using labelled datasets. Different machine learning problems such as classification and regression were explained using practical examples.

Several commonly used algorithms were discussed including:

- Linear Regression
- Logistic Regression
- Decision Trees
- K-Nearest Neighbor (KNN)
- Support Vector Machine (SVM)

Each algorithm has different strengths depending on the type of problem being solved.

After learning the algorithms, I studied Model Evaluation and Optimization. Developing a machine learning model is only the first step. Its performance must also be evaluated using various statistical metrics.

Important evaluation parameters studied included:

- Accuracy
- Precision
- Recall

- F1 Score
- Confusion Matrix

The importance of avoiding overfitting and underfitting was also discussed. Various optimization techniques such as train-test splitting, cross-validation, and parameter tuning were introduced.

By the end of Week 2, I gained a clear understanding of how machine learning models are developed, trained, tested, and evaluated.

Learning Outcomes of Week 2

- Machine Learning fundamentals
 - Supervised Learning
 - Classification and Regression
 - Model Evaluation
 - Performance Metrics
 - Model Optimization
-

4.3 Week 3 – Deep Learning and Convolutional Neural Networks

Week 3 introduced **Deep Learning**, which is an advanced area of Artificial Intelligence inspired by the structure and functioning of the human brain.

Deep Learning models consist of multiple interconnected neural networks capable of learning complex patterns from large amounts of data. These models have revolutionized fields such as image recognition, speech processing, language translation, and autonomous vehicles.

The internship explained the structure of an **Artificial Neural Network (ANN)**, including:

- Input Layer
- Hidden Layers
- Output Layer

Each neuron processes information received from previous neurons and passes the output forward through activation functions.

The next module focused on **Convolutional Neural Networks (CNNs)**, which are specialized deep learning models designed for image processing.

CNNs automatically identify important features from images without requiring manual feature extraction. They are widely used in:

- Image Classification
- Face Recognition
- Medical Image Analysis
- Traffic Sign Detection
- Object Detection

The internship also demonstrated the working of CNNs for image classification using sample datasets. I learned how convolution layers, pooling layers, activation functions, and fully connected layers work together to classify images accurately.

This module provided practical insight into one of the most popular deep learning architectures used in modern Artificial Intelligence.

Learning Outcomes of Week 3

- Deep Learning concepts
- Artificial Neural Networks
- CNN Architecture
- Image Classification
- Feature Extraction
- Image Recognition

4.4 Week 4 – Advanced Artificial Intelligence Applications

The final week covered advanced applications of Artificial Intelligence with emphasis on **Natural Language Processing (NLP)** and **Reinforcement Learning**.

Natural Language Processing enables computers to understand, interpret, and generate human language. It combines Artificial Intelligence with linguistics to enable communication between humans and machines.

The internship introduced several NLP tasks including:

- Text Processing
- Tokenization
- Sentiment Analysis
- Text Classification

- Language Translation
- Chatbots

Practical implementation was demonstrated using Python libraries such as **NLTK** and **SpaCy**.

The second part of Week 4 focused on Reinforcement Learning.

Unlike supervised learning, reinforcement learning enables an intelligent agent to learn through interaction with its environment by receiving rewards or penalties for its actions.

Important concepts studied included:

- Agent
- Environment
- State
- Action
- Reward
- Policy

The internship also introduced Q-Learning as one of the fundamental reinforcement learning algorithms.

Finally, different evaluation metrics used in Artificial Intelligence were discussed, including Accuracy, Precision, Recall, and F1 Score, helping students understand how AI models are validated before deployment.

Learning Outcomes of Week 4

- Natural Language Processing
- Text Analytics
- NLTK
- SpaCy
- Reinforcement Learning
- Q-Learning
- AI Model Evaluation

CHAPTER 5

TECHNOLOGIES USED

During the internship, various software tools, programming languages, and libraries were used for learning Artificial Intelligence concepts and implementing practical exercises.

Technology	Purpose
Python	Primary programming language
Google Colab	Cloud-based coding environment
Jupyter Notebook	Interactive Python notebooks
NumPy	Numerical computation
Pandas	Data manipulation
Matplotlib	Data visualization
Scikit-learn	Machine Learning algorithms
TensorFlow	Deep Learning framework
Keras	Neural Network development
NLTK	Natural Language Processing
SpaCy	Advanced NLP
OpenCV	Image processing

These technologies provided practical exposure to modern Artificial Intelligence development and helped me understand how AI applications are developed in professional environments.

CHAPTER 6

WEEKLY LEARNING ACTIVITIES

Week 1

- Studied Artificial Intelligence fundamentals.
 - Learned Python programming basics.
 - Understood AI applications in industry.
 - Performed data preprocessing exercises.
 - Created simple data visualizations.
-

Week 2

- Learned Machine Learning concepts.
 - Understood supervised learning algorithms.
 - Trained basic machine learning models.
 - Studied evaluation metrics.
 - Explored optimization techniques.
-

Week 3

- Learned Deep Learning fundamentals.
 - Studied Artificial Neural Networks.
 - Understood Convolutional Neural Networks.
 - Explored image classification techniques.
 - Practiced CNN model implementation.
-

Week 4

- Learned Natural Language Processing.
- Implemented text preprocessing techniques.
- Studied Reinforcement Learning concepts.

- Understood Q-Learning.
- Completed final quizzes and practical learning modules.
- Successfully completed all **21 learning modules** and achieved **100% course completion**, reinforcing my understanding of Artificial Intelligence concepts from fundamentals to advanced applications.

Summary of Internship Learning

The internship followed a structured learning path that gradually introduced Artificial Intelligence concepts from basic programming to advanced AI applications. Each week built upon the previous week's concepts, making it easier to understand modern AI technologies. By completing all training modules, I developed a strong theoretical foundation in Artificial Intelligence along with practical knowledge of machine learning, deep learning, computer vision, natural language processing, and reinforcement learning. This experience has increased my confidence in pursuing more advanced AI projects and further studies in this domain.

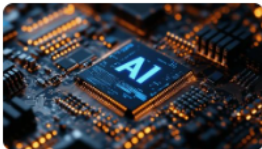
Hi, Dev Vishwakarma 🌟

DV

📖 Enrolled (1) ▾

🔖 Wishlist

🧩 Quiz Attempts



1-month Internship

1-Month Artificial Intelligence Internship

21 of 21 lessons • 100% Complete

CHAPTER 7

PROJECTS STUDIED DURING THE INTERNSHIP

As part of the Artificial Intelligence internship, various real-world project ideas were introduced to help interns understand the practical applications of Artificial Intelligence and Machine Learning. Although the internship primarily focused on learning AI concepts and understanding their implementation, these projects provided valuable insight into how AI techniques can be used to solve real-world problems.

The following projects were studied during the internship.

Project 1

Handwritten Digit Recognition using Convolutional Neural Networks (CNN)

7.1 Introduction

Handwritten Digit Recognition is one of the most common beginner projects in Deep Learning. It involves developing an Artificial Intelligence model capable of recognizing handwritten numerical digits (0–9) from grayscale images.

This project generally uses the **MNIST (Modified National Institute of Standards and Technology)** dataset, which contains thousands of handwritten digit images collected from different individuals. Each image is labelled with its corresponding digit, making it suitable for supervised learning.

The project demonstrates how Convolutional Neural Networks (CNNs) can automatically extract image features and classify handwritten digits with high accuracy.

During the internship, this project was introduced to understand the workflow of image classification using Deep Learning techniques.

7.2 Objective

The main objectives of this project are:

- To understand image classification using Deep Learning.
- To study the architecture of Convolutional Neural Networks.
- To understand feature extraction from images.

- To explore the practical use of TensorFlow and Keras.
 - To gain knowledge of real-world computer vision applications.
-

7.3 Problem Statement

Human handwriting varies significantly from person to person. Traditional programming methods cannot easily recognize different handwriting styles.

Artificial Intelligence solves this problem by training Convolutional Neural Networks on thousands of labelled handwritten images. After sufficient training, the model learns various patterns and becomes capable of predicting unknown handwritten digits accurately.

7.5 Technologies Required

- Python
 - TensorFlow
 - Keras
 - NumPy
 - Matplotlib
 - Google Colab / Jupyter Notebook
-

7.6 Expected Working

The system accepts a handwritten digit image as input.

The image is processed and normalized before being passed through multiple convolution and pooling layers. These layers automatically identify important features from the image.

The extracted features are then passed through fully connected neural network layers that classify the image into one of the ten numerical classes (0–9).

7.7 Applications

- Bank cheque processing
- Postal code recognition
- Automatic form processing
- Educational software
- Digital handwriting recognition

- Document digitization
-

7.8 Learning Outcome

Through studying this project, I gained an understanding of:

- Image classification
- CNN architecture
- Feature extraction
- Neural networks
- Computer Vision fundamentals

Project 2

Speech-to-Text Transcription System

7.9 Introduction

Speech Recognition is one of the fastest-growing applications of Artificial Intelligence. It enables computers to convert spoken language into readable text automatically.

This technology is widely used in voice assistants, online meetings, customer support systems, smart devices, and accessibility tools.

During the internship, the Speech-to-Text project was introduced to demonstrate how Artificial Intelligence processes human speech using Natural Language Processing (NLP) techniques.

7.10 Objective

The objectives of this project are:

- To understand Speech Recognition technology.
 - To study Natural Language Processing concepts.
 - To learn audio preprocessing.
 - To understand Python speech recognition libraries.
 - To explore voice-enabled AI applications.
-

7.11 Problem Statement

Typing long text manually can be time-consuming.

Speech Recognition systems reduce this effort by converting spoken audio into written text automatically using Artificial Intelligence algorithms.

Such systems improve accessibility and increase productivity across multiple domains.

7.13 Technologies Required

- Python
- SpeechRecognition Library
- PyAudio
- Google Speech API

- Google Colab
 - VS Code
-

7.14 Expected Working

The system records human speech through a microphone.

The recorded audio is processed and converted into text using Speech Recognition libraries.

The generated text is displayed to the user for further editing or storage.

7.15 Applications

- Voice Assistants
 - Live Captioning
 - Customer Service
 - Healthcare
 - Smart Home Devices
 - Accessibility Support
 - Virtual Meetings
-

7.16 Learning Outcome

After studying this project, I understood:

- Audio processing fundamentals
- Speech Recognition workflow
- Natural Language Processing basics
- AI-based voice applications
- Python libraries used in speech recognition

CHAPTER 8

SKILLS ACQUIRED

The internship significantly enhanced both my technical and professional skills.

Technical Skills

- Artificial Intelligence Fundamentals
- Python Programming
- Machine Learning Concepts
- Data Preprocessing
- Data Visualization
- Deep Learning Basics
- CNN Fundamentals
- Natural Language Processing
- Reinforcement Learning Concepts
- Model Evaluation Techniques

Soft Skills

- Problem Solving
- Logical Thinking
- Time Management
- Self-Learning
- Technical Documentation
- Analytical Thinking

CHAPTER 9

CHALLENGES FACED

During the internship, I encountered several learning challenges while studying Artificial Intelligence concepts.

Initially, understanding Machine Learning algorithms and their mathematical intuition required additional practice. The large number of AI libraries and frameworks also made it challenging to identify the appropriate tool for different tasks.

Deep Learning concepts such as neural networks and convolutional neural networks were new to me, requiring repeated study and revision. Understanding evaluation metrics like Precision, Recall, and F1-Score also required careful practice.

Since the internship was conducted in online mode, maintaining consistency and self-discipline while completing learning modules was another challenge. However, by following the structured course material and revising concepts regularly, I was able to overcome these difficulties.

These challenges improved my analytical thinking and strengthened my interest in Artificial Intelligence.

CHAPTER 10

LEARNING OUTCOMES

The internship provided valuable exposure to Artificial Intelligence and significantly improved my understanding of modern AI technologies.

By the end of the internship, I had achieved the following learning outcomes:

- Understood the fundamentals of Artificial Intelligence.
- Learned Python programming for AI applications.
- Studied Machine Learning algorithms.
- Understood data preprocessing techniques.
- Learned model evaluation methods.
- Gained knowledge of Deep Learning concepts.
- Explored Convolutional Neural Networks.
- Understood Natural Language Processing basics.
- Learned introductory Reinforcement Learning concepts.
- Developed confidence to explore advanced AI projects independently.

CHAPTER 11

CONCLUSION

The one-month Artificial Intelligence Internship at Codec Technologies was a valuable learning experience that strengthened my understanding of Artificial Intelligence and its practical applications. The internship provided structured learning modules covering Python programming, Machine Learning, Deep Learning, Convolutional Neural Networks, Natural Language Processing, and Reinforcement Learning.

Although the internship primarily focused on conceptual learning and understanding AI workflows, it introduced me to practical project ideas such as Handwritten Digit Recognition and Speech-to-Text Transcription. These projects helped me understand how Artificial Intelligence techniques are applied to solve real-world problems and inspired me to explore them further through future implementation.

The knowledge and skills gained during this internship have enhanced my confidence in pursuing more advanced topics in Artificial Intelligence. I believe this experience has provided a strong foundation that will support my academic growth and future career in Computer Science and Artificial Intelligence.

REFERENCES

1. Python Software Foundation. *Python Documentation*. <https://docs.python.org/>
2. TensorFlow Developers. *TensorFlow Documentation*. <https://www.tensorflow.org/>
3. Scikit-learn Developers. *Scikit-learn User Guide*. <https://scikit-learn.org/>
4. NumPy Documentation. <https://numpy.org/>
5. Pandas Documentation. <https://pandas.pydata.org/>
6. Keras Documentation. <https://keras.io/>
7. NLTK Documentation. <https://www.nltk.org/>
8. Google Colab Documentation. <https://colab.research.google.com/>
9. Artificial Intelligence learning modules provided during the Codec Technologies Internship.